

Workshop 2

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1. A spherical balloon with radius r has volume $V = \frac{4}{3}\pi r^3$. Find a function that represents the amount of air required to inflate the balloon from a radius of r to a radius of $r + 1$.
2. An open rectangular box with volume $2m^3$ has a square base. Express the surface area of the box as a function of the length of a side of the base.
3. Let g be the function defined as $g(x) = -3x + 8$ and $f(x)$ be the linear function such that $f(-3) = 4$ and $f(-1) = 2$. Write as an interval the set

$$A = \{x \in \mathbb{R} : g(x) < f(x)\}$$

4. Let $f(x) = \frac{-20x + 4}{kx + 10}$. Find $k \in \mathbb{R}$ such that the line $L : x = 2$ is a vertical asymptote of f .

5. Evaluate the difference quotient for the given function. Simplify your answer.

$$(a) \quad f(x) = 4 + 3x - x^3, \quad \frac{f(3+h) - f(3)}{h}$$

$$(b) \quad f(x) = x^3, \quad \frac{f(a+h) - f(a)}{h}$$

$$(c) \quad f(x) = \frac{1}{x}, \quad \frac{f(x) - f(a)}{x - a}$$

What is this quotient representing?

6. Consider the following function:

$$f(x) = \frac{2}{\sqrt{x^2 - 1}}$$

- (a) Find its maximal possible domain and image.
 - (b) Is it injective?
 - (c) If it is injective; then, find its inverse onto its image.
 - (d) If it is not injective; then, find another domain where it is injective and find its inverse onto its domain here.
7. Answer the all the question of the previous problem, but for this little change of the the last function:

$$f(x) = \frac{2}{x^2 - 1}$$

Did you get the same answers?

8. Find the quadratic function f such that

- (a) The x -position of the vertex is $x_v = 2$
- (b) The $\text{Im}(f) = [5, +\infty)$
- (c) $f(4) = 13$

9. Find the domain of the following functions

$$(a) \quad f(x) = \frac{1}{1 - \tan(x)}$$

10. Find the domain of the following function

$$f(x) = \frac{1 - e^{x^2}}{1 - e^{1-x^2}}$$

11. Let g be the real function defined as

$$g(x) = |x^2 - 1| - |x^2 - 4|$$

Is g odd, even or nothing? It's injective?

12. Solve each equation for x .

(a) $e^{7-4x} = 6$

(b) $\ln(x^2 - 1) = 3$

(c) $e^{2x} - 3e^x + 2 = 0$

(d) $\ln(\ln(x)) = 1$

13. Find $f + g$, $f \cdot g$ and $f \circ g$ and state their domain.

(a) $f(x) = x^3 + 2x^2$, $g(x) = 3x^2 - 1$

14. Prove that $\cos(\sin^{-1}(x)) = \sqrt{1 - x^2}$

15. Find the functions $f \circ g$, $g \circ f$, $f \circ f$ and $g \circ g$ and their domains.

(a) $f(x) = 3x + 5$, $g(x) = x^2 + x$

(b) $f(x) = \sqrt{x+1}$, $g(x) = 4x - 3$

16. Find the inverse of the following functions

(a) $f(x) = 1 + \sqrt{2 + 3x}$

(b) $f(x) = \frac{4x - 1}{2x + 3}$

17. Decide if the following statement is true or false

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ to be an even function. If $f(x)$ is increasing in $A_+ = [0, \infty)$ then it is decreasing in $A_- = (-\infty, 0]$

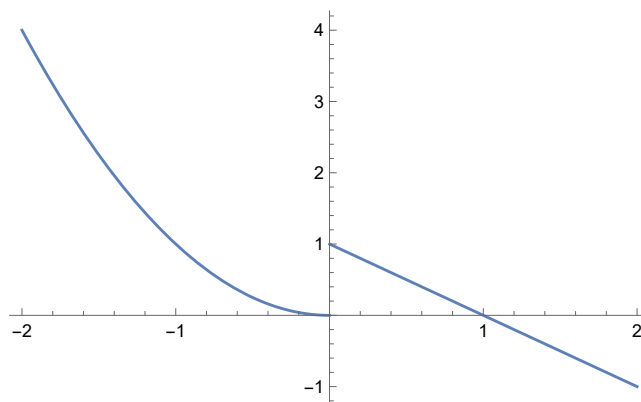
If it's true prove it and if it's false give a counterexample.

18. Decide if the following statement is true or false

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ to be an odd function. If $f(x)$ is increasing in $A_+ = [0, \infty)$ then it is increasing in $A_- = (-\infty, 0]$

If it's true prove it and if it's false give a counterexample.

19. Determine the increasing and decreasing intervals of the following function.



Here, f is given by

$$f(x) = \begin{cases} x^2, & \text{if } x \leq 0 \\ 1 - x, & \text{otherwise} \end{cases}$$